

SMARTSHARE PROJECT – INTERVIEW EXPLANATION GUIDE

PROJECT NAME SmartShare – Bucket-Based File Sharing Platform (Ongoing)

1. PROJECT OVERVIEW

SmartShare is a full-stack Android application being developed to provide organized file sharing and storage using a bucket-based approach. Instead of storing files in traditional folders, users create buckets for different purposes such as academics, projects, or teams and manage files within them.

The frontend is developed using Kotlin and Jetpack Compose. The backend is planned using FastAPI and PostgreSQL. The system supports user authentication, bucket management, file upload/download, and controlled sharing through share codes or links.

This project combines Android development, backend development, database design, API design, networking concepts, and software architecture.

2. PROBLEM STATEMENT

Students and small teams often need a simple way to organize and share files.

Current solutions: - Google Drive - Dropbox - OneDrive

Issues: - Cluttered organization - Not project-oriented - Sharing can become difficult for temporary collaboration

Solution: SmartShare introduces buckets that logically organize files and make sharing easier.

3. CORE IDEA

Instead of:

Folder |— notes.pdf |— report.docx

Use:

DBMS Bucket |— notes.pdf |— assignment.pdf

Networking Bucket |— report.docx |— topology.png

Each bucket acts as an independent storage container.

4. TECHNOLOGY STACK

Frontend: - Kotlin - Jetpack Compose - Android Studio

Backend: - FastAPI

Database: - PostgreSQL

Architecture: - MVVM - REST APIs

5. HIGH LEVEL ARCHITECTURE

Android App ↓ REST API ↓ FastAPI Backend ↓ PostgreSQL Database ↓ File Storage

6. RESPONSIBILITY OF EACH COMPONENT

Android App: - Login - Registration - Bucket Management - File Management - Settings

FastAPI: - Authentication - Business Logic - Validation - Database Access

PostgreSQL: - Users - Buckets - Files Metadata - Sharing Information

File Storage: - Actual files - PDFs - ZIP files - Images - Videos

7. DATABASE DESIGN

users

id username email password_hash created_at

buckets

id name owner_id created_at

files

id bucket_id filename filepath size uploaded_at

shares

id bucket_id share_code expiry_time

8. API DESIGN

Authentication: POST /register POST /login

Buckets: GET /buckets POST /bucket PUT /bucket/{id} DELETE /bucket/{id}

Files: GET /files POST /upload DELETE /file/{id}

Sharing: POST /share GET /share/{code}

9. LOGIN FLOW

User enters: Email Password

Android sends:

```
{ "email": "rohit@gmail.com", "password": "mypassword" }
```

POST /login

Backend: - Validate User - Verify Password - Generate Token

Response:

```
{ "token": "jwt-token" }
```

10. REGISTRATION FLOW

Android sends:

```
{ "username": "rohit", "email": "rohit@gmail.com", "password": "mypassword" }
```

Backend: - Validate Data - Hash Password - Store User

Success:

```
{ "message": "Registration successful" }
```

Failure:

```
{ "error": "User already exists" }
```

11. FILE UPLOAD FLOW

User selects file

Android ↓ POST /upload ↓ FastAPI ↓ Store File ↓ Insert Metadata into PostgreSQL

Response:

```
{ "message": "Upload successful" }
```

12. SECURITY

Passwords are never stored directly.

Backend: - Hashes passwords - Stores password_hash

Database never stores plain-text passwords.

13. WHY THESE TECHNOLOGIES

Why Kotlin? - Official Android language - Null safety - Concise syntax - Compose support

Why FastAPI? - Lightweight - Fast - Automatic API documentation - Easy integration with PostgreSQL

Why PostgreSQL? - Reliable - Supports relationships - Production-ready database

14. CURRENT STATUS

The project is currently under development.

Completed: - Requirements analysis - Feature planning - High-level architecture - Technology stack selection - Database design - API planning - Android development environment setup - Kotlin and Jetpack Compose learning - Initial Android UI development

In Progress: - Android frontend development - Navigation implementation - Authentication workflow

Planned: - Backend development - Database integration - File upload/download - Bucket sharing system

15. ONE-MINUTE INTERVIEW ANSWER

SmartShare is a full-stack Android file sharing platform that organizes files using bucket-based storage. Users can create buckets, upload files, and share them with others using controlled sharing mechanisms. The frontend is built using Kotlin and Jetpack Compose, while the backend uses FastAPI and PostgreSQL. The project helps me learn Android development, REST APIs, database design, authentication, and full-stack software architecture.